Jose Esparza

Location: Trujillo, La Libertad – Peru **Phone:** +51 997 189 179
E-mail: joseesparzadc@gmail.com **GitHub:** pebeto
ORCID iD: 0000-0002-9372-3763 **LinkedIn:** josesparza

Senior AI Engineer | **LLM / RAG Systems** | **Open-source Julia**

Experience

- **Senior Artificial Intelligence Engineer @ Subatomic** July 2026 — Present
  Building Subatomic's agentic AI-workforce platform (FastAPI + Next.js, Supabase + pgvector, Redis, Google Cloud Run): a LangGraph agent system with a GPT-4.1 ReAct supervisor, a self-critiquing four-sub-agent reasoning graph, and 10+ specialized agents integrating GitHub, Confluence, Monday.com, Slack, and Google Drive. Shipped a production RAG pipeline (hybrid vector + full-text search, RRF/MMR reranking, LLM-driven chunk-level RBAC) with OpenAI-to-Gemini failover; engineered a Redis job queue and WebSocket notifications powering an interaction-capture engine and a closed-loop prospect-signal system feeding the CRM.
- **Senior Machine Learning Engineer (Contract) @ Maximo** March 2026 — Present
  Architected and shipped a production agentic banking assistant for WhatsApp customer support: FastAPI + LangGraph agent, RAG over a Postgres + pgvector knowledge base, AWS-backed OTP delivery (SES/SNS), audio transcription via faster-whisper, and an LLM-generated monthly financial-report subsystem on EventBridge cron. Built an MCP (Model Context Protocol) server with FastMCP exposing internal tools to LLM agents; designed Metabase BI dashboards modeling fintech revenue (interchange, GMV, spreads) and user-growth indicators.
- **Senior Software Engineer (ML/LLM Systems) @ The New England Center for Children** March 2023 — March 2026
  Built a transformer-based recommender that surfaces semantically related lessons across the curriculum (e.g., a "tomato" query returns not only direct lessons but adjacent skills like grocery shopping and burger preparation), with sub-second response. Led end-to-end deployment of a fine-tuned Llama 3.1 8B (LLaMA-Factory + LoRA, three rank iterations) for automated curriculum-resource generation, grounded via a RAG layer over NECC's internal PDF technical documentation; expert reviewers confirmed exact match with the curriculum's English voice.
- **Senior Data Consultant @ Sankio del Peru** September 2022 — June 2025
  Multi-discipline consulting (data science, data engineering, analytics) for Peruvian clients across banking, logistics, and food services. Built classical ML models for client-specific use cases, orchestrated production ETL pipelines, and delivered BI dashboards driving stakeholder decisions.
- **Lead Data Engineer @ Teamcore** July 2022 — January 2023
  Built AWS-based ETL pipelines (Glue, Athena, Lambda, S3) ingesting product-availability data from major retailer B2B platforms — Cencosud, Falabella, Oxxo, Soriana, Walmart. Led a team of five engineers and coordinated with Product on client requirements.

Earlier Experience

- **Backend Developer / Data Scientist @ Apurata** Nov 2021 — Jul 2022
- **Full-stack Engineer @ Barrick Gold Corporation** May 2021 — Nov 2021
- **Research Intern @ Binghamton University** Jan 2021 — Aug 2021
- **HPC Semi-Senior Developer @ Everis** Feb 2020 — Dec 2020
- **Full-stack Developer @ DAVOS SALUD** Jul 2019 — Dec 2019

Skills

- **LLM & ML Systems**: PyTorch, TensorFlow, Hugging Face Transformers, vLLM, llm-compressor, LLaMA-Factory, Unsloth, LangChain, LangGraph, LlamaIndex, MCP (Model Context Protocol / FastMCP), OpenAI API, Google Gemini API, scikit-learn, NumPy, Pandas, CUDA, faster-whisper
- **Data & Cloud**: Spark/PySpark, Airflow, Prefect, Databricks (Delta Lake, MLflow, Feature Store, Model Serving), AWS (SageMaker, Glue, Redshift, Athena, Lambda, S3, EC2, DynamoDB), GCP (BigQuery, VertexAI, Dataflow, Pub/Sub, Cloud Run, Looker), Supabase
- **Databases**: PostgreSQL (pgvector), MSSQL, MySQL, MongoDB, Redis, Valkey, Neo4j, OracleDB
- **Backend & Web**: FastAPI, Flask, Django, Node.js, TypeScript, React.js, Next.js
- **Languages & Tooling**: Python, SQL, TypeScript, Julia, C++, Java, Bash; Docker, Kubernetes, Git, GitHub Actions, Jenkins
- **Observability**: Prometheus, Grafana, Datadog, Sentry, Langfuse, LangSmith

Education

- **MSc. Computer Science**
  University of London, September 2023 — Expected December 2026
- **B.S. Software Engineering** — GPA 4.0
  Universidad Privada del Norte, March 2016 — December 2020

Publications

- **Esparza, J.** (2026). A Review of Topic Modeling: From Probabilistic Foundations to Neural and Multimodal Approaches. Preprint, Zenodo. **doi:** 10.5281/zenodo.20280741
- **Esparza, J.**, Bejarano, G., Ramesh, A., & Seetharam, A. (2021). Understanding the Issues Surrounding COVID-19 Vaccine Roll Out via User Tweets. In *Computational Data and Social Networks (CSoNet 2021)*, Lecture Notes in Computer Science, pp. 197–205. Springer. **doi:** 10.1007/978-3-030-91434-9_18
- **Esparza, J.** (2020). Convolutional Neural Networks in Respiratory Conditions Detection: A Systematic Review of the Last 4 Years. Research poster, Summer School of Machine Learning, Skoltech.

Research and Projects

- **LatamGPT-70B-NVFP4: NVFP4 (W4A4) quantization of LatamGPT's 70B model on a single 32GB-RAM workstation.**
  Personal — Open-source model release
  June 2026 — **Model**
- **Amigo: Privacy-first, on-device voice companion running a small LLM fully on a laptop CPU.**
  Hugging Face Build Small Hackathon — Open-source project
  June 2026 — faster-whisper + Qwen3.5-4B (llama.cpp) + Piper TTS + Chroma RAG + persona LoRA — **Model**
- **MichiBoost.jl: Pure Julia gradient boosting with native categorical feature handling and symmetric decision trees.**
  Personal — Open-source project
  Ongoing project — **Source**
- **DearDiary.jl: A machine learning experiment tracking tool for Julia**
  JuliaAI - Open-source project
  Ongoing project — **Source**
- **MLJ.jl: A Julia machine learning framework**
  JuliaAI - Open-source project
  Ongoing project — **Source**
- **Empowering Julia-based Data Science with MLflow**
  Google Summer of Code — Open-source project
  May 2023 — September 2023 – **Source**

Honors & Awards

- **Top 10% Spot, SMILES 2020**
  Skolkovo Institute of Science and Technology — August 2020
  Awarded an exclusive presentation spot (top 10% of 2,000+ applicants) to present the work *"Convolutional Neural Networks in Respiratory Conditions Detection: A Systematic Review of the Last 4 Years"*.
- **REPU Computer Science Selection**
  Research Experience for Peruvian Undergraduates — October 2020
  Selected as a REPU Computer Science undergraduate participant, working under the mentorship of Gisella Bejarano in Professor Arti Ramesh's research lab at Binghamton University.
- **OAS Cybersecurity Scholarship Recipient**
  Organization of American States — May 2019
  Awarded a spot in the *"Creando una Trayectoria Profesional en Seguridad Digital"* program, receiving specialized training at Universidad Federico Villarreal in Lima.

Academic Service

- **Judge, REPU Seminar 2026**
  Research Experience for Peruvian Undergraduates — April 2026
  Evaluated undergraduate research presentations across scientific disciplines and provided constructive feedback to participants.
- **Reviewer, IEEE Intercon 2024**
  IEEE — July 2024
  Reviewed academic paper submissions for technical accuracy, originality, and alignment with conference themes.

Courses & Certifications

- **Deep Learning with Flux.jl**
  JuliaAcademy
  February 2023 — Certificate No. cert_my5hq86p
- **Deep Learning Specialization**
  deeplearning.ai
  September 2020 — Certificate
- **Machine Learning**
  Stanford University
  November 2019 — Certificate